COURSE OUTLINE

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Device Health Monitoring Automation with Claude

From VIP, HA and version checks to the right ServiceNow ticket, automatically

3 days (8 hours per day) | Intermediate | Instructor-led, hands-on labs | Capstone project

About this course

Who should attend

Network operations, load balancer and data centre engineers, NOC leads, ITSM and automation teams running F5, NetScaler, Radware, other load balancers (Anaya LB) and Cisco ACI.

Prerequisites

- Working knowledge of load balancers (VIPs, pools, HA pairs, config sync)
- Basic REST API and JSON concepts
- Basic Python and Git
- Familiarity with ServiceNow incidents and change requests is helpful, not required

What you will be able to do

- Define a platform-neutral health model covering VIP status, uptime, HA state, sync state, version and last change for every device class
- Collect health signals from F5 iControl REST and tmsh, Citrix NetScaler NITRO, Radware Alteon via Cyber Controller, other load balancers (Anaya LB) and Cisco APIC REST
- Use Claude with tool use and structured JSON outputs to turn raw evidence into a cited, severity-ranked health verdict
- Create or update ServiceNow incidents through the Table API and Flow Designer with deduplication and correlation IDs
- Run the whole workflow on a schedule under approvals, guardrails and a complete audit trail

Tools and technologies

Claude Code and Agent SDK | Model Context Protocol | Python | Pydantic

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F5 iControl REST / tmsh | NetScaler NITRO API | Radware Alteon / Cyber Controller | Cisco APIC REST API | ServiceNow Table API / Flow Designer | GitHub | GitHub Actions | pytest

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DAY 1 — 4 modules

MODULE 1

What device health means across the fleet

- The six health signals: VIP status, uptime, HA status, sync status when in HA, software version and last change performed
- How each signal looks on F5 BIG-IP, Citrix NetScaler, Radware Alteon, other load balancers (Anaya LB) and Cisco ACI
- Why manual morning checks fail: missed sync drift, stale tickets, duplicate incidents
- The pattern for the whole course: Claude plans and explains, deterministic code executes, a person approves, everything is logged

HANDS-ON LAB
Walk the lab fleet by hand with tmsh, NITRO and the APIC GUI and record what a healthy baseline looks like for each platform.

DELIVERABLE
Health signal matrix per platform.

MODULE 2

Setting up Claude Code, the Agent SDK and MCP for this workflow

- Installing Claude Code, project CLAUDE.md conventions and the Agent SDK in Python
- Building a read-only MCP server that exposes get_vip_status, get_ha_state and get_version tools over the lab devices
- Tool use and structured JSON outputs: defining a HealthReading schema with Pydantic
- Secrets handling for device and ServiceNow credentials outside the model context

HANDS-ON LAB
Scaffold the project with Claude Code, run the MCP server locally and have the agent read the version of lb-f5-01 through a tool call.

DELIVERABLE
Working agent project with a read-only device MCP server.

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MODULE 3

Collecting health from F5 and NetScaler

- F5 iControl REST: /mgmt/tm/ltm/virtual/stats for VIP availability, /mgmt/tm/cm/device and /mgmt/tm/cm/sync-status for HA and sync, /mgmt/tm/sys/version
- tmsh equivalents (show sys failover, show cm sync-status, show sys version) for devices without REST access
- NetScaler NITRO: /nitro/v1/stat/lbvserver, /nitro/v1/config/hanode, /nitro/v1/config/nsversion and system uptime
- Normalising every response into the shared HealthReading schema with retries and timeouts

HANDS-ON LAB
Write the F5 and NetScaler collectors with Claude Code, inspect every diff, and run them against the lab HA pairs.

DELIVERABLE
F5 and NetScaler collectors with pytest fixtures.

MODULE 4

Collecting health from Radware, other load balancers (Anaya LB) and Cisco ACI

- Radware Alteon health through Cyber Controller REST and SNMP: virtual server state, VRRP peer status, sync status, image version
- A generic adapter interface for other load balancers (Anaya LB) and devices that only offer CLI or SNMP
- Cisco APIC REST API: topSystem and fabricNode for uptime and role, firmwareRunning for version, fabricHealthTotal for fabric health
- Handling partial data: marking a signal as unknown rather than guessing

HANDS-ON LAB
Add the Alteon, generic and APIC collectors and prove a missing signal surfaces as unknown, not healthy.

DELIVERABLE
Complete multi-platform collector library.

DAY 2 4 modules

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MODULE 5

Last change performed: correlating config history and change records

- F5 audit log via tmsh show sys log audit (or /var/log/audit) and sync events in /var/log/ltm, NetScaler ns.log and config save timestamps, APIC audit log via the aaaModLR class
- Pulling closed and scheduled change requests from the ServiceNow change_request table for each configuration item
- Git-tracked configuration snapshots and diffs as a second source of truth
- Building a timeline that links a health change to the last change performed on the device

HANDS-ON LAB

Make a controlled change on lb-ns-02, then watch the pipeline find it in the device log, the Git snapshot and the ServiceNow change record.

DELIVERABLE

Change timeline per device.

MODULE 6

Claude as the triage analyst: structured health verdicts

- A HealthVerdict schema: severity, affected signals, cited evidence, likely cause, recommended action, confidence
- Prompting with evidence bundles and forbidding conclusions without a cited reading
- Severity rules as code: sync drift in an HA pair is never below major, a single VIP down on a standby is informational
- Testing the analyst with contradictory and missing evidence

HANDS-ON LAB

Build the triage agent with the Agent SDK, feed it twenty synthetic evidence bundles and score its verdicts against the rules.

DELIVERABLE

Triage agent with verdict test suite.

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MODULE 7

Creating or updating ServiceNow tickets

- ServiceNow Table API: incident create, query by correlation_id and cmdb_ci, update work notes and state
- Decision logic: update the open ticket, reopen a resolved one, or create new, and never create duplicates
- Flow Designer flows that assign, notify and set priority from the verdict severity
- Idempotent runs and dry-run mode that prints the ticket payload without sending it

HANDS-ON LAB
Run the pipeline twice against a failing HA pair and prove it creates one incident, then updates it with new work notes.

DELIVERABLE
ServiceNow ticket adapter with dedup logic.

MODULE 8

Guardrails, approvals and audit for health monitoring

- Read-only by default: the MCP server exposes no write tools to devices
- Approval gates for consequential actions: ticket priority P1 and P2, reopening resolved tickets, paging on-call
- Audit record for every run: readings, prompt, verdict, ticket action, approver and timestamps in SQLite
- Rate limits, timeouts and circuit breakers so a flapping device cannot flood ServiceNow

HANDS-ON LAB
Add the approval gate and audit log, then trigger a P1 verdict and watch the run pause until a person approves.

DELIVERABLE
Approval and audit layer.

DAY 3 3 modules

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MODULE 9

Scheduling and running the workflow with GitHub Actions

- Repository layout, pull requests for every collector or rule change and required reviews
- GitHub Actions on a schedule and on demand, with secrets and a self-hosted runner that can reach the devices
- Run summaries as job artifacts and a daily health digest posted back to ServiceNow
- Handling device maintenance windows and planned downtime without false tickets

HANDS-ON LAB
Ship the pipeline through a pull request, run it from Actions on a schedule and suppress alerts during a declared maintenance window.

DELIVERABLE
Scheduled, reviewed and versioned monitoring pipeline.

MODULE 10

Testing, dashboards and operations

- pytest suites for collectors, severity rules and ticket decision logic with recorded API fixtures
- Exporting health readings to Prometheus and a Grafana fleet dashboard beside the ticket flow
- Operator runbook: what to do when a collector fails, when a verdict is wrong, when ServiceNow is down
- Measuring the outcome: time to ticket, duplicate rate, false positives

HANDS-ON LAB
Break a collector on purpose, watch the dashboard and the audit log explain it, and write the runbook entry.

DELIVERABLE
Test suite, dashboard and operator runbook.

MODULE 11

Capstone: end-to-end device health monitoring for a mixed fleet

- Fleet in, verdicts out, with the right ServiceNow ticket created or updated and nothing consequential without approval
- Scheduled GitHub Actions run against the full lab fleet with the approval gate and audit log exercised end to end
- Dry-run and live run compared, with duplicate-ticket and false-positive checks against the acceptance criteria
- Limitations statement, operator handover and team demo

HANDS-ON LAB
Capstone build and demo on the full lab fleet.

DELIVERABLE
End-to-end monitoring platform and runbook.

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Capstone

Capstone project

A scheduled platform that collects VIP, uptime, HA, sync, version and last-change evidence from F5, NetScaler, Radware, other load balancers (Anaya LB) and Cisco ACI, lets Claude produce a cited health verdict, and creates or updates the correct ServiceNow ticket with approvals and a full audit trail.

Assessment

Assessment area	Weight	Evidence
Working functionality	25%	Demonstrable end-to-end workflow in a repeatable lab
Security and safety controls	20%	Threat model, authorization boundaries, approval gates, safe failure
Testing and evaluation	20%	Automated tests, golden dataset, measured quality and known limitations
Engineering quality	15%	Clear interfaces, error handling, reproducible setup, maintainable code
Evidence and explainability	10%	Traceable findings, sources, assumptions, logs
Documentation and presentation	10%	Architecture diagram, README, runbook, demo, lessons learned

Engineering principles used throughout

- Model output is a proposal, not a fact: every finding and generated change is validated by deterministic checks and tests.
- Authorization lives outside the model: tool access is enforced by application code, scoped credentials and policy.
- Human approval for consequential actions: deploying a mitigation, changing production configuration, rotating credentials.
- Decisions, evidence, model and tool versions, approvals and outcomes are logged without recording secrets.
- A model adapter and a tool adapter layer keep every build provider-agnostic and enterprise-portable.
- Quality is measured: false positives, false negatives, reliability, latency, cost and safe failure, not just a demo.

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